

Road Traffic Accidents

Ethiopia · 2018–2023

- Overview
- Causes & Risk
- Driver & Vehicle
- Time & Place
- Collision & Road

FILTERS

Year

2018

2019

2020

2021

2022

2023

Severity

All

RECORDS

12,316

FATAL RATE

1.3%

Slight Serious Fatal

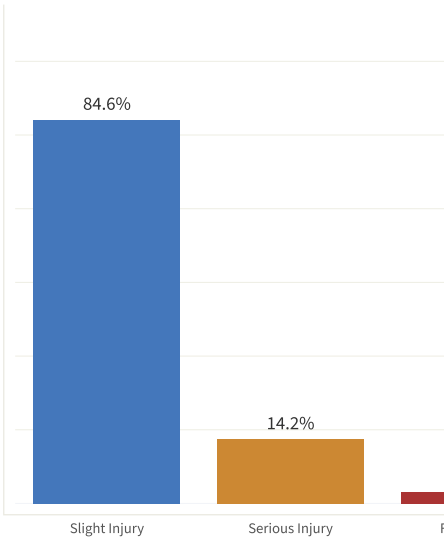
12,316 total incidents · 5 pages

Overview

Scale, severity distribution, yearly trend, and escalation funnel — Ethiopia 2018–2023.

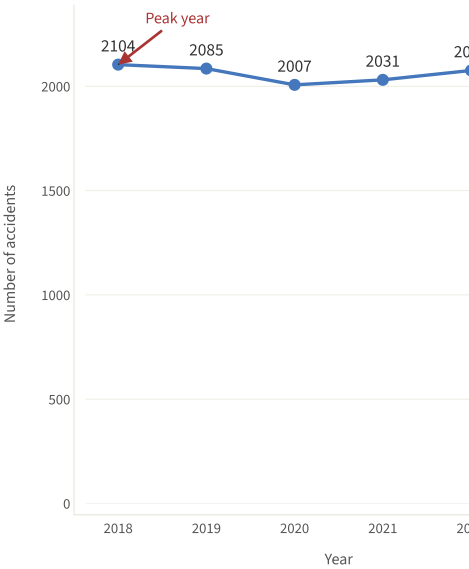
TOTAL ACCIDENTS 12,316	SLIGHT INJURIES 10,415	SERIOUS INJURIES 1,743	FATAL CASES 158 (1.3%)
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Accidents by severity level



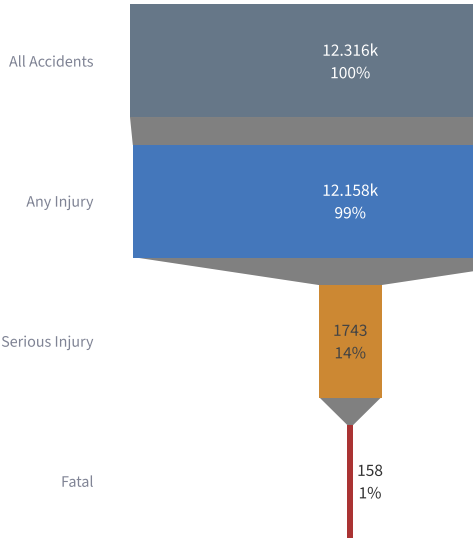
Slight injuries make up 84.6% of all accidents. Fatal cases are 1.3% — small in count, highest in consequence.

Accidents recorded per year

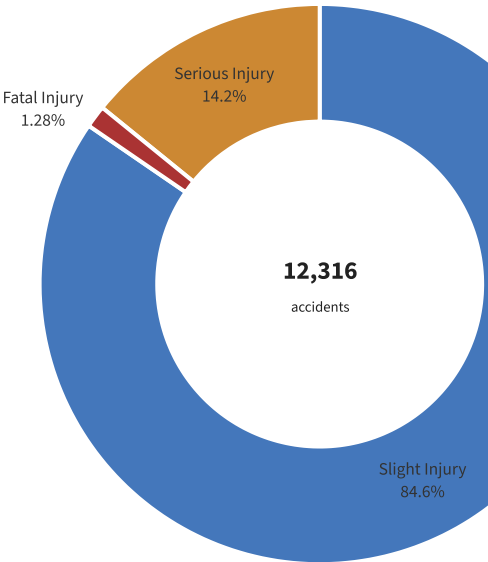


Annual totals hold steady near 2,000. Serious injuries show a gradual rise post-2020.

Accident severity escalation funnel



Severity breakdown



Each stage filters down to the next severity level. Fatal cases represent 1.3% of all accidents — the critical intervention target.

The donut confirms the scale: with 12,316 accidents, 84.6% are slight — but the 1.3% fatal share is where policy must focus.

Ethiopia Road Traffic Accidents · 2018–2023 · 12,316 records · Streamlit & Plotly

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Causes & Risk

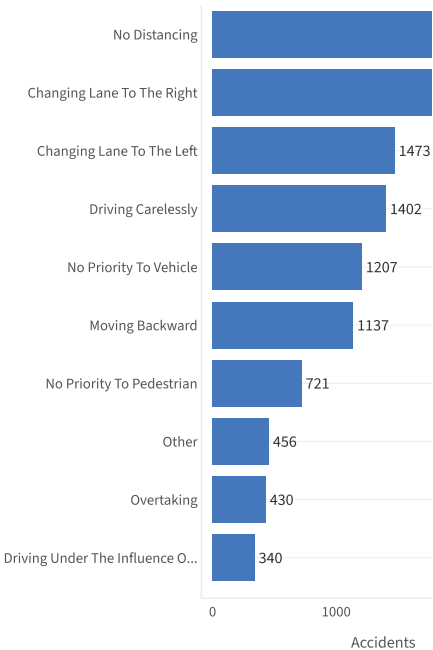
Which causes are most frequent, which are most deadly, and how cause volumes shift year by year.

ACCIDENTS
12,316

MOST COMMON CAUSE
No Distancing

FATAL RATE
1.3%

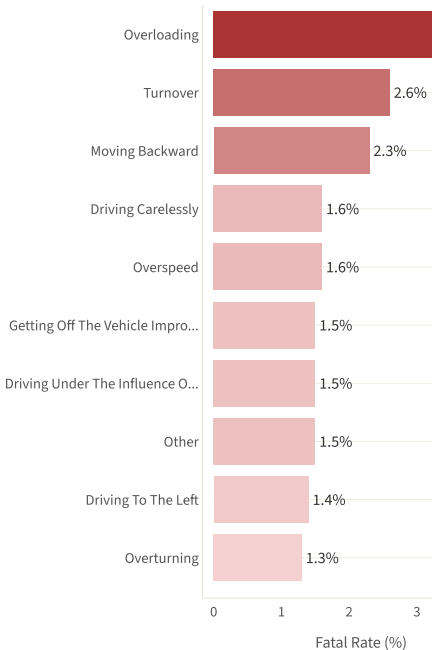
Top 10 most frequent causes



No distancing leads by a wide margin — a behavioural cause directly addressable through enforcement.

Deadliest causes by fatal rate

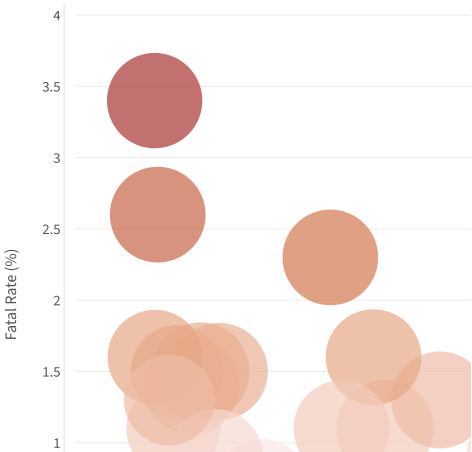
Causes with fewer than 30 records excluded



Overloading and turnover carry fatal rates above 2.5% — nearly double the overall 1.3% average.

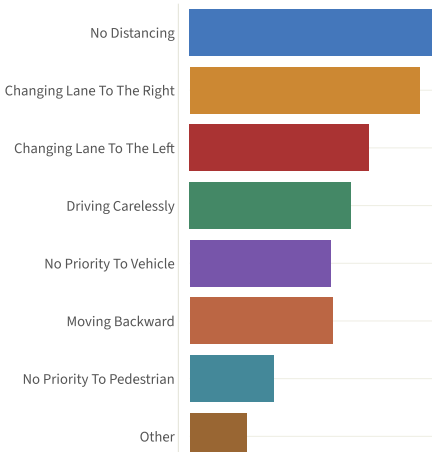
Frequency vs fatal risk

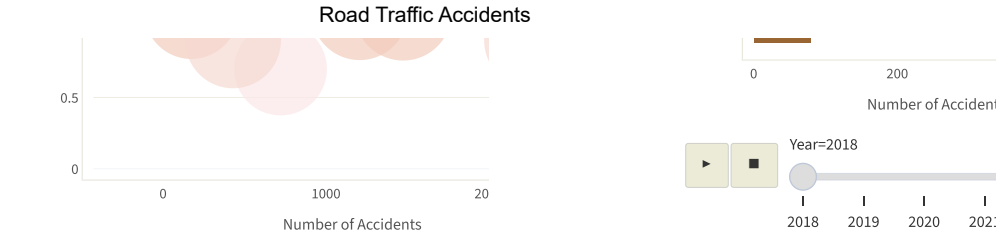
Bubble size = avg casualties per accident — a 3-variable view of cause danger



Top causes by year

Animation · press ► · Fixed axis and colors — bar length changes are real year-to-year shifts.





Top-right quadrant = frequent AND deadly. Moving backward and driving carelessly sit dangerously high on both axes.

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Driver & Vehicle

Who is involved in accidents — age, experience, and vehicle type — and how risk changes over time.

Filter by gender

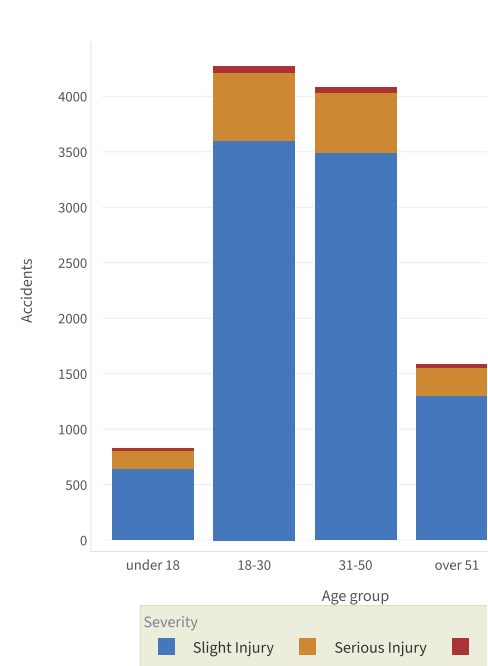
All

ACCIDENTS
12,316

MOST COMMON OUTCOME
Slight Injury

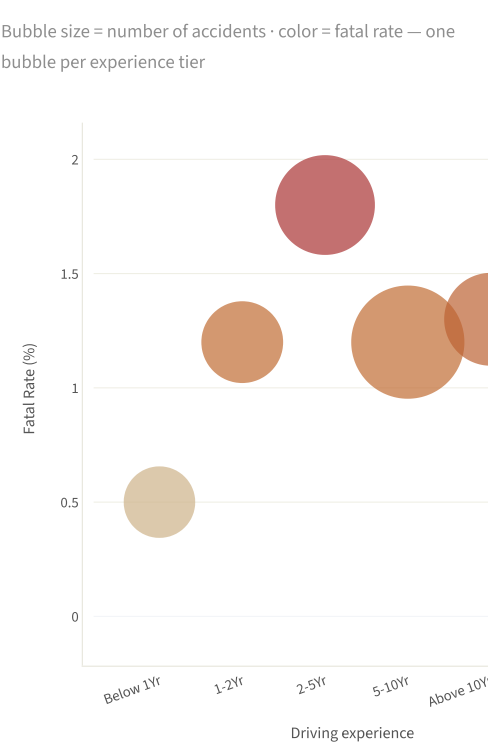
FATAL RATE
1.3%

Driver age group by severity



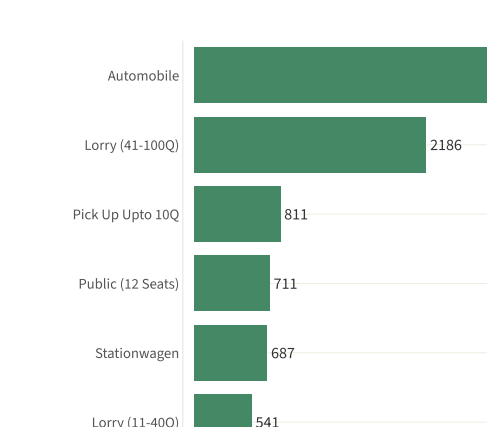
18–30 and 31–50 dominate by volume. Over-51 drivers show a proportionally higher serious and fatal share.

Driving experience — accidents vs fatal risk

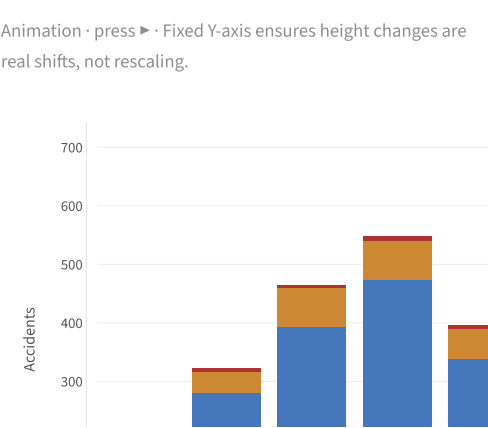


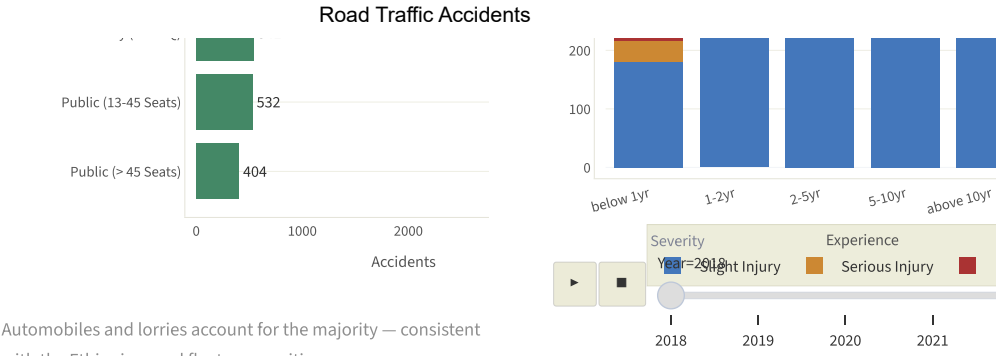
Bubble size = accident frequency. Height = fatal risk. 2-5yr drivers have the most accidents; above-10yr retains surprisingly high fatal rate.

Vehicle types involved in accidents



Driving experience by severity over time





2-5yr and 5-10yr groups dominate across all years. The severity split reveals fatal risk persists even with experience.

Driver risk summary table

Experience	Accidents	Fatal Cases	Serious Cases	Fatal Rate (%)
below 1yr	1342	7	207	
1-2yr	1756	21	228	
2-5yr	2613	46	381	
5-10yr	3363	41	462	
above 10yr	2262	29	323	
no licence	118	0	13	

Table confirms that 2-5yr drivers have the highest absolute accident count, while above-10yr drivers retain a disproportionate fatal rate.

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FATAL RATE

1.3%

Slight **Serious** **Fatal**

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Time & Place

When accidents cluster through the day and which area types carry the highest fatal risk.

ACCIDENTS

12,316

PEAK ACCIDENT HOUR

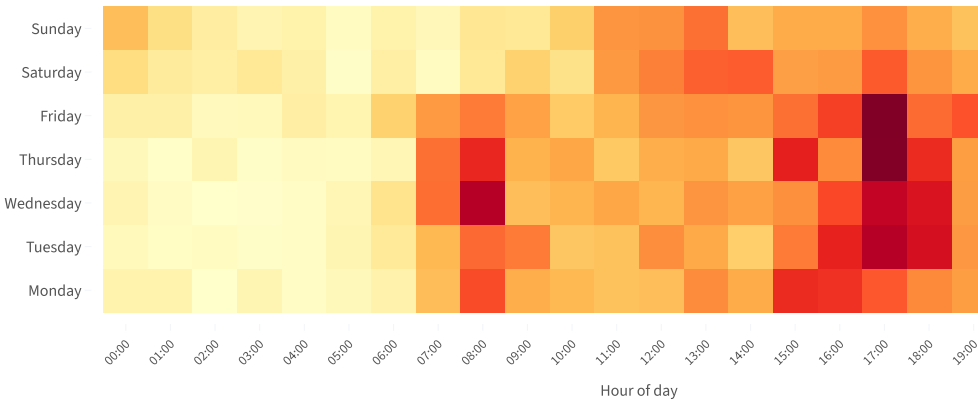
17:00

FATAL RATE

1.3%

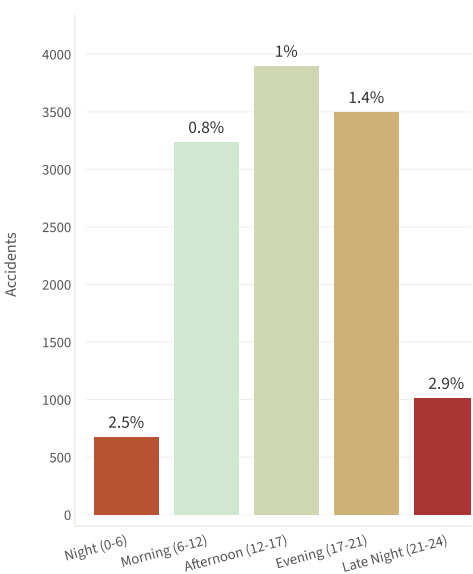
Accident heatmap — hour of day vs day of week

Darker cells = more accidents. The evening rush (16–19h) stands out across all weekdays.

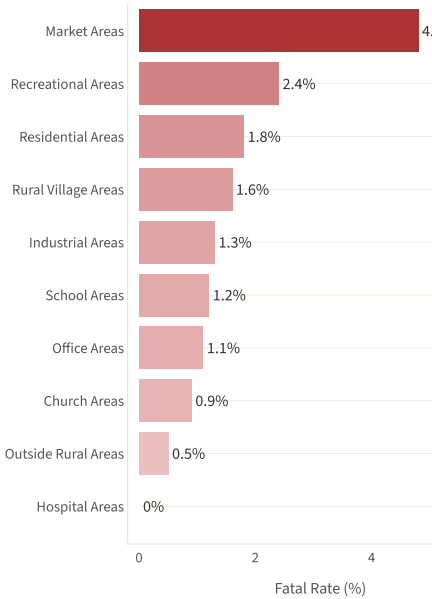


Accidents by time of day

Bar height = accident count · bar color = fatal rate — two variables, one chart



Fatal rate by area type

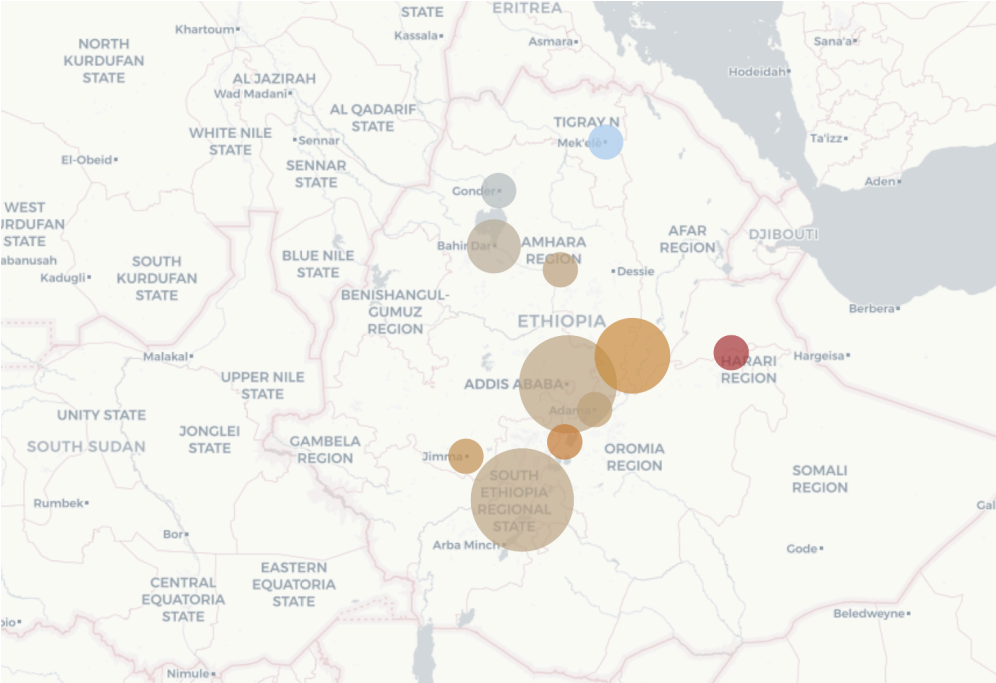


Market and recreational areas carry the highest fatal rates — pedestrian exposure and mixed traffic conditions.

Afternoon has the most accidents. Night and late night have 2-3x higher fatal rates despite far fewer incidents.

Accident area types — mapped across Ethiopia

The dataset records area category, not GPS coordinates. Each bubble is placed at a real Ethiopian city chosen to represent that area type (e.g. market areas → Dire Dawa · hospital areas → Mekelle · church areas → Bahir Dar). All cities are at least 76 km apart — no overlap. Bubble size = accident count · Colour = fatal rate.



Large bubbles = high accident volume. Red colour = high fatal rate. Market areas (Dire Dawa) are small in count but the most dangerous — 4.8% fatal rate, the highest of any area type.

Area	City	Accidents	Fatal Rate (%)
Market Areas	Dire Dawa	63	4.8
Recreational Areas	Ziway	328	2.4
Residential Areas	Haramaya	2060	1.8
Rural Village Areas	Jimma	64	1.6
Industrial Areas	Adama	456	1.3
School Areas	Debre Berhan	415	1.2
Office Areas	Addis Ababa	3451	1.1
Other	Wolaita Sodo	3819	1.1
Church Areas	Bahir Dar	1060	0.9
Outside Rural Areas	Gondar	218	0.5

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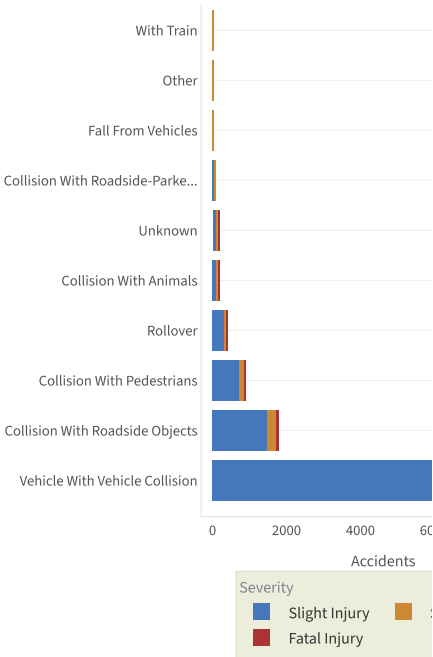
How accidents happen — collision type, road surface, who gets hurt, and cause risk evolution over time.

ACCIDENTS
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MOST COMMON COLLISION
Vehicle With Vehicle Collis...

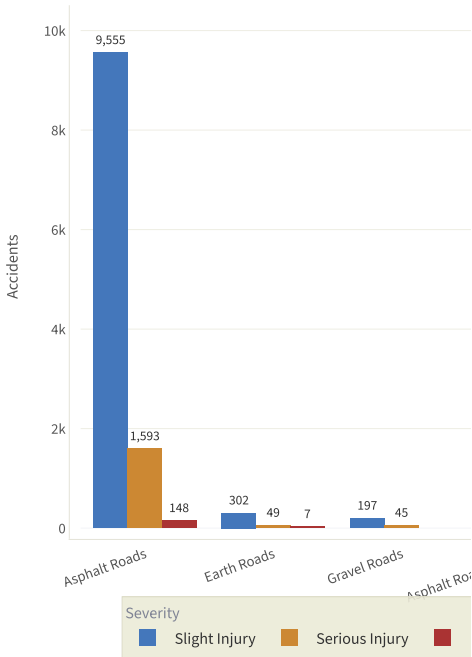
FATAL RATE
1.3%

Accidents by collision type and severity



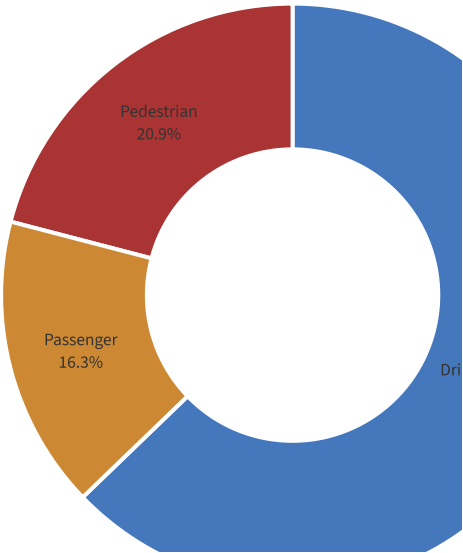
Vehicle-to-vehicle collisions dominate. Pedestrian collisions are rarer but carry a disproportionately high fatal share.

Accidents by road surface and severity



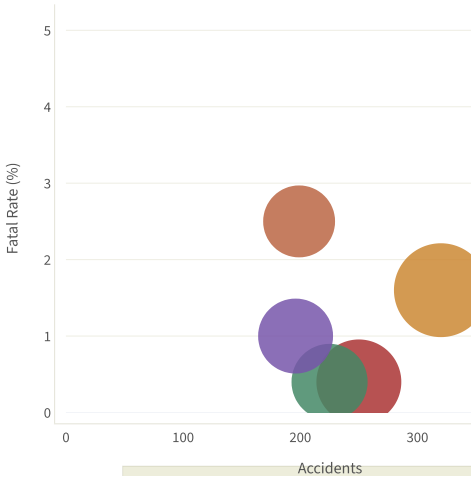
Grouped bars make it easy to compare severity within each road type. Earth roads have a much higher fatal proportion relative to their count.

Who is being hurt — casualty class



Cause risk evolution by year

Animation · press ▶ · X = frequency · Y = fatal rate · size = total casualties · Fixed axes & colors.



Road Traffic Accidents

Drivers and riders make up 62% of casualties. Pedestrians at 21% are a significant and particularly vulnerable group.



Moving right = more accidents. Moving up = higher fatal rate.
Growing bubble = more total casualties. Same color per cause across all years.

Road & collision risk summary

Road Surface	Accidents	Fatal Cases	Fatal Rate (%)
earth roads	358	7	2.0
asphalt roads	11296	148	1.3
asphalt roads with some distress	81	1	1.2
gravel roads	242	0	0.0

Earth roads have the highest fatal rate despite lower accident count — high-risk surfaces require targeted intervention.